STA2023 Exam 2 — SIDE A — THE PRACTICE EXAM, SOLVED (the real exam copies it)

■ GREEN = the question ■ RED = rule / answer ■ BLACK = the work ■ BLUE = every way they can ask it

Formula → plug in → arithmetic → box it. Probabilities: 4 decimals (0.4 → 0.4000); μ, σ, x: 2 decimals. No work = no credit. Blue tables = every other way a question can be phrased; the Q4, Q6, Q7 tables are on side B.

Q1 (6 pts) — Two-way table: or / and / given

"The table below summarizes the employment status of 78 residents." Males / Females: Full-time 22 / 18 (row 40), Part-time 6 / 10 (16), Unemployed 3 / 2 (5), Not in labor force 5 / 12 (17); males 36, females 42, total 78.

First: total each row and column at the table's edge (if the counts come in words, draw the table yourself). Every answer is then **a count ÷ the group you are picking from**.

(a) probability that a randomly selected resident is male? one category (a row or a column) → its total ÷ grand total (78)

$$P(\text{male}) = \frac{36}{78} = 0.4615$$

(b) …is part-time? same rule: add the row's cells, ÷ grand total

$$P(\text{part-time}) = \frac{6+10}{78} = \frac{16}{78} = 0.2051$$

(c) …is male OR unemployed? OR: $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$ — adding counts the overlap (male AND unemployed = 3) twice, so take it out once

$$P(\text{M or U}) = \frac{36}{78} + \frac{5}{78} - \frac{3}{78} = \frac{38}{78} = 0.4872$$

Watch out: forgetting the overlap gives 41/78 × (check: 36 males + 2 unemployed women = 38 ✓)

(d) …is female AND not in the labor force (NLF)? AND in a table = the ONE cell where the row and column meet, ÷ grand total

$$P(\text{F and NLF}) = \frac{12}{78} = 0.1538$$

(e) …is part-time GIVEN that the resident is female? GIVEN: $P(B \mid A) = \frac{\#(A \text{ and } B)}{\#(A)}$ — the letter after | is the given group (the word after "given"): divide by its total, 42, not 78

$$P(\text{PT} \mid \text{Female}) = \frac{10}{42} = 0.2381$$

Watch out: 10/78 = 0.1282 is the classic error.

EVERY WAY THEY CAN ASK A TABLE QUESTION — worked (males 36, females 42, N = 78)

is male (a column)	column total ÷ grand total: $\frac{36}{78}$ = **0.4615**
is part-time (a row)	row total ÷ grand total: $\frac{6+10}{78} = \frac{16}{78}$ = **0.2051**
is NOT male	$1 - \frac{36}{78} = \frac{42}{78}$ = **0.5385**
male OR unemployed	add, take the overlap (3) out once: $\frac{36}{78} + \frac{5}{78} - \frac{3}{78} = \frac{38}{78}$ = **0.4872**
female AND not in labor force	the one cell: $\frac{12}{78}$ = **0.1538**
part-time GIVEN female	÷ the given group (42 females): $\frac{10}{42}$ = **0.2381**
female GIVEN part-time	other direction, ÷ the 16 part-timers: $\frac{10}{16}$ = **0.6250**
full-time OR part-time (same variable)	can't both happen → just add: $\frac{40}{78} + \frac{16}{78} = \frac{56}{78}$ = **0.7179**
part-time and female independent?	P(PT | F) = 0.2381 vs P(PT) = 0.2051 → not equal → **dependent**
two residents picked, both male (no replacement)	$\frac{36}{78} \cdot \frac{35}{77}$ (2nd pick: one fewer male, one fewer person) = **0.2098**
two residents, at least one male (no replacement)	1 − P(neither male) = $1 - \frac{42}{78} \cdot \frac{41}{77} = 1 - 0.2867$ = **0.7133**

Q2 (2 pts) — Valid discrete probability distribution?

"Determine whether the following distribution represents a valid discrete probability distribution. If not, explain why." x: 0, 1, 2, 3, 4 with P(x): 0.34, 0.21, 0.13, 0.04, 0.01

Valid only if (1) every $0 \le P(x) \le 1$ AND (2) $\sum P(x) = 1$. Write BOTH checks (given decimals must add to exactly 1).

Rule 1: 0.34, 0.21, 0.13, 0.04, 0.01 are all between 0 and 1 ✓

Rule 2: add them:

$$0.34 + 0.21 + 0.13 + 0.04 + 0.01 = 0.73 \neq 1 \ \times$$

NOT a valid probability distribution — the probabilities add to 0.73, not 1.

EVERY WAY THEY CAN ASK IT — worked

the P(x) add to more or less than 1	$0.34 + 0.21 + 0.13 + 0.04 + 0.01 = 0.73 \neq 1$ → **NOT valid**
a P(x) below 0 or above 1	e.g. P(3) = −0.10 → rule 1 fails even if the sum is 1 → **NOT valid**
all from 0 to 1 (0 and 1 allowed) and they add to exactly 1	e.g. $0.2 + 0.4 + 0.3 + 0.1 = 1$ → both rules pass → **valid**
find the missing P(x)	1 − (the others): $1 - (0.34 + 0.21 + 0.13 + 0.04) = 1 - 0.72$ = **0.2800**

Q3 (8 pts) — Stanley Cup: discrete random variable

"The following data represent the number of games played, X, in the Stanley Cup before a champion was determined." x: 4, 5, 6, 7 with frequency 20, 20, 26, 17.

(a) construct the probability distribution

$P(x) = \frac{\text{frequency}}{\text{total}}$ **(frequency = how many times that x happened) total = 20 + 20 + 26 + 17 = 83**

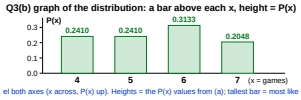
$$P(4) = \frac{20}{83} = 0.2410 \qquad P(5) = \frac{20}{83} = 0.2410$$

$$P(6) = \frac{26}{83} = 0.3133 \qquad P(7) = \frac{17}{83} = 0.2048$$

Check: 0.2410 + 0.2410 + 0.3133 + 0.2048 = 1.0001 ≈ 1 ✓ (rounding).

(b) graph the discrete probability distribution

x across, P(x) up; a bar above each x with height P(x); label both axes. In words: "bars at 4, 5, 6, 7 with heights 0.2410, 0.2410, 0.3133, 0.2048; tallest at x = 6."



(c) compute and interpret the mean of X

$\mu = \sum x \cdot P(x)$ **— a weighted average: each x counts as much as its probability**

$$\mu = 4(0.2410) + 5(0.2410) + 6(0.3133) + 7(0.2048)$$
$$= 0.9640 + 1.2050 + 1.8798 + 1.4336 = 5.48 \text{ games}$$

μ ≈ 5.48 games **Interpretation (part of the points):** over many Stanley Cup series, the average length is about 5.48 games.

(d) compute the standard deviation of X

$\sigma = \sqrt{\sum (x-\mu)^2 \cdot P(x)}$ **with μ = 5.48: subtract, square, × P(x), add, √**

(4 − 5.48)² = 2.1904 → × 0.2410 = 0.5279 · (5 − 5.48)² = 0.2304 → × 0.2410 = 0.0555 · (6 − 5.48)² = 0.2704 → × 0.3133 = 0.0847 · (7 − 5.48)² = 2.3104 → × 0.2048 = 0.4732

$$\sigma^2 = 0.5279 + 0.0555 + 0.0847 + 0.4732 = 1.1413$$
$$\sigma = \sqrt{1.1413} = 1.07 \text{ games}$$

σ ≈ 1.07 games

Watch out: don't stop at the variance; no ÷ 4 or ÷ 83 here.

EVERY WAY THEY CAN ASK IT — worked (P(4) = 0.2410, P(5) = 0.2410, P(6) = 0.3133, P(7) = 0.2048)

at least / at most / "or more" / "or less" keep the number IN; more than / fewer than leave it OUT.

exactly 6	P(6) = **0.3133**
at least 6 = 6 or more	x = 6, 7: $0.3133 + 0.2048$ = **0.5181**
more than 6 = 7 or more	x = 7: P(7) = **0.2048**
fewer than 6 = 5 or less	x = 4, 5: $0.2410 + 0.2410$ = **0.4820**
at most 5 = 5 or less	same list x = 4, 5: $0.2410 + 0.2410$ = **0.4820**
between 5 and 6 inclusive	x = 5, 6: $0.2410 + 0.3133$ = **0.5543**
not 4 (= at least 5)	$1 - P(4) = 1 - 0.2410$ = **0.7590**
at most 6 (= fewer than 7, = not 7)	$1 - P(7) = 1 - 0.2048$ = **0.7952**
mode · expected value · variance · SD	mode = the biggest P(x) → 6 · E(X) = the mean = 5.48 · σ² = 1.1413 (no √) · $\sigma = \sqrt{1.1413}$ = **1.07**
is a 7-game series unusual?	P(7) = 0.2048 is not below 0.05; also $\mu \pm 2\sigma = 5.48 \pm 2.14$ = 3.34 to 7.62 and 7 is inside → **not unusual**

Q4 (8 pts) — Driving age: binomial

"According to a Gallup poll, 60% of U.S. women 18 years old or older stated that the minimum driving age should be 18. In a random sample of 10 U.S. women 18 years old or older, find the probability that:"

Binomial: n = the sample size (10 yes/no trials); p = the percent in the question as a decimal. **n = 10, p = 0.60, 1 − p = 0.40, X = number who say yes.**

$$P(x) = {}_nC_x\, p^x (1-p)^{n-x} \qquad {}_nC_x = \frac{n!}{x!\,(n-x)!}$$

(a) exactly 3 believe the minimum driving age should be 18

$${}_{10}C_3 = \frac{10!}{3!\,7!} = \frac{10 \cdot 9 \cdot 8}{3 \cdot 2 \cdot 1} = 120$$
$$P(3) = {}_{10}C_3\,(0.60)^3(0.40)^7 = 120(0.216)(0.0016384) = 0.0425$$

0.0425

Watch out: the power on 0.40 is n − x = 7, not 3. ${}_{10}C_3$: 10! = 10·9·8·7!, the 7! cancels → 10·9·8 ÷ (3·2·1), or press the nCr key.

(b) fewer than 3 believe… "fewer than 3" = 0, 1, 2 — the 3 is OUT. Add the three ($_{10}C_0$ = 1, 0.6^0 = 1); round each to 4 decimals, then add those.

$$P(0) = {}_{10}C_0(0.6)^0(0.4)^{10} = 1(1)(0.0001049) = 0.0001$$
$$P(1) = {}_{10}C_1(0.6)^1(0.4)^9 = 10(0.6)(0.0002621) = 0.0016$$
$$P(2) = {}_{10}C_2(0.6)^2(0.4)^8 = 45(0.36)(0.0006554) = 0.0106$$
$$P(X < 3) = 0.0001 + 0.0016 + 0.0106 = 0.0123$$

0.0123

(c) between 1 and 3 (inclusive) believe… inclusive → x = 1, 2, 3. Reuse (a) and (b) — write "from part (a)".

$$P(1 \le X \le 3) = P(1) + P(2) + P(3) = 0.0016 + 0.0106 + 0.0425 = 0.0547$$

0.0547

(d) sample of 200: expected number and standard deviation binomial shortcuts: $\mu = np$, $\sigma = \sqrt{np(1-p)}$

$$\mu = np = 200(0.60) = 120$$
$$\sigma = \sqrt{np(1-p)} = \sqrt{200(0.60)(0.40)} = \sqrt{48} = 6.93$$

μ = 120, σ ≈ 6.93 (48 is the variance — take the √)

Q5 (5 pts) — Daily temperature: uniform

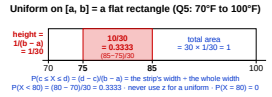
"The daytime temperature X (°F) in a coastal city during summer is uniformly distributed between 70°F and 100°F."

a = smaller end 70, b = bigger end 100, width 30. $\text{height} = \frac{1}{b-a}$ $P(c \le X \le d) = \frac{d-c}{b-a}$ **= the piece's share of the width. "more than c" → (b − c)/(b − a); "less than d" → (d − a)/(b − a). No z (that's Q6).**

(a) sketch / describe the pdf (= the probability graph)

$$\text{height} = \frac{1}{b-a} = \frac{1}{100-70} = \frac{1}{30} \qquad \text{area} = 30 \cdot \frac{1}{30} = 1 \ \checkmark$$

a flat line (rectangle) at height 1/30 from 70 to 100, and 0 everywhere else



(b) P(75 ≤ X ≤ 85)

$$P(75 \le X \le 85) = \frac{85-75}{100-70} = \frac{10}{30} = 0.3333$$

0.3333

(c) P(X < 80)

$$P(X < 80) = \frac{80-70}{100-70} = \frac{10}{30} = 0.3333$$

0.3333

EVERY WAY THEY CAN ASK IT — worked (uniform from 70 to 100, width 30)

height of the pdf	$\frac{1}{100-70} = \frac{1}{30}$ = **0.0333**
between 75 and 85	$\frac{85-75}{30} = \frac{10}{30}$ = **0.3333**
below 80 (same as at most 80)	$\frac{80-70}{30} = \frac{10}{30}$ = **0.3333**
above 90 (same as at least 90)	$\frac{100-90}{30} = \frac{10}{30}$ = **0.3333**
at least 95	$\frac{100-95}{30} = \frac{5}{30}$ = **0.1667**
exactly 80	a single point has no width → **0.0000**
between 60 and 75 (60 is outside)	only 70 to 75 is inside: $\frac{75-70}{30} = \frac{5}{30}$ = **0.1667**
20% of days are cooler than what?	$x = 70 + 0.20(30) = 70 + 6$ = **76°F**
the hottest 10% start where?	top 10% → 90% below it: $x = 70 + (1 - 0.10)(30) = 70 + 27$ = **97°F**

Q6 (6 pts) — Standard normal: the exam's z hint (= the table entry) is the area to the LEFT

"Use the standard normal table (on the real exam: the printed z hints). Round final answers to four decimal places."

The exam prints the table values as hints. LEFT (below) = the hint · RIGHT (above) = 1 − hint · BETWEEN = (left of the bigger z) − (left of the smaller z); bigger = further right · $z = \frac{x-\mu}{\sigma}$ to 2 decimals

(a) area to the left of z = −1.10

the hint (or table row −1.1, column .00): **0.1357** (a hint for a negative z is already the left area — use it as is)

(b) area between z = −0.60 and z = 1.45

hints: left of 1.45 = 0.9265; left of −0.60 = 0.2743.

$$\text{between} = 0.9265 - 0.2743 = 0.6522$$

0.6522

Watch out: subtract the AREAS, never the z's; a negative answer means you subtracted the wrong way.

(c) μ = 80, σ = 10: P(70 < X < 95)

$$z_1 = \frac{70-80}{10} = -1.00 \ \rightarrow \ \text{left} = 0.1587$$
$$z_2 = \frac{95-80}{10} = 1.50 \ \rightarrow \ \text{left} = 0.9332$$
$$P(70 < X < 95) = 0.9332 - 0.1587 = 0.7745$$

0.7745

Q7 (8 pts) — Wechsler IQ: normal with μ = 100, σ = 15

"The Wechsler Intelligence Scale for Children is approximately normally distributed with mean μ = 100 and standard deviation σ = 15."

Forward (a)–(c): $z = \frac{x-100}{15}$ → hint → below: left · above: 1 − left · between: bigger left − smaller left. Backward (d): percent → left area → the hint's z → $x = \mu + z\sigma$ (x to 1 decimal).

(a) probability a test taker scores above 125

$$z = \frac{125-100}{15} = 1.67 \ \rightarrow \ \text{left} = 0.9525$$
$$P(X > 125) = 1 - 0.9525 = 0.0475$$

0.0475

Watch out: 0.9525 is the #1 error; fewer than half score above 125 → the answer must be below 0.5.

(b) probability a test taker scores below 90

$$z = \frac{90-100}{15} = -0.67$$
$$P(X < 90) = \text{left of } -0.67 = 0.2514$$

0.2514 (keep the minus sign on z)

(c) proportion scoring between 110 and 140

$$z_1 = \frac{110-100}{15} = 0.67 \ \rightarrow \ \text{left} = 0.7486$$
$$z_2 = \frac{140-100}{15} = 2.67 \ \rightarrow \ \text{left} = 0.9962$$
$$P(110 < X < 140) = 0.9962 - 0.7486 = 0.2476$$

0.2476 (proportion = probability; 4 decimals)

(d) score at the 98th percentile

98th percentile = 98% score below it → area to the LEFT = 0.98 → the exam's hint gives z = 2.05 (from a table: 0.9798 is the entry closest to 0.98).

$$x = \mu + z\sigma = 100 + 2.05(15)$$
$$= 100 + 30.75 = 130.75 \approx 131$$

≈ 131 (130.75)

Watch out: never put 0.98 into the z formula — it is an area, not a score.

STA2023 Exam 2 — SIDE B — RULES, FORMULAS, PICTURES

■ BLUE = what it means ■ **RED = formula / rule** ■ GREEN = example ■ BLACK = steps

SYMBOLS — what they mean

Symbol	Plain English
Σ	"add them all up"
μ ("mu") / x̄ ("x-bar")	the mean (average): μ population, x̄ sample
σ ("sigma") / s	standard deviation (SD) = typical gap from the mean; σ^2, s^2 = variance = SD^2
P(E), E^c, N(E)	probability of event E (0 = impossible … 1 = certain); E^c = "not E"; N(E) = # of outcomes in E, N(S) = # of all outcomes
X / x, P(x)	X = the random variable (the number an experiment produces); x = one value; P(x) = the chance X comes out equal to x; μ_X, σ_X = its mean and SD; E(X) = expected value = μ_X
P(F | E); z	probability of F **given** E happened; z = how many SDs a value sits above (+) or below (−) the mean
n, p; ${}_nC_r$, ${}_nP_r$, n!	binomial trials and P(success); C = pick r of n, order ignored; P = order counts; n! = n·(n−1)·…·1

μ, σ or capital N → whole population; x̄, s or small n → a sample. ≤ / ≥ = at most / at least (the end number counts); ≠ = not equal. Probabilities: 4 decimals (0.4 → 0.4000); μ, σ, x: 2.

CH 4 — PROBABILITY

- **Probability** = the long-run fraction of times something happens (Law of Large Numbers: more trials → the fraction settles at the probability). **0 ≤ P(E) ≤ 1; all outcomes together add to 1. Unusual event** = probability below 0.05.
- **Sample space S** = everything that could happen; an **event** = any part of it.
- **Classical** (equally likely outcomes — dice, cards, coins): count. $P(E) = \frac{N(E)}{N(S)}$ P(two heads) = 1/4; P(at least one tail) = 3/4.
- **Empirical** (from data): $P(E) \approx \frac{\text{frequency of } E}{\text{number of trials}}$ 22 of 200 carpool → 0.11 (interpret: "if many people were selected, about 11 of every 100 would carpool"); 5 walk → 5/200 = 0.025 < 0.05 → unusual. **Subjective** = educated guess.
- **Disjoint** (mutually exclusive) = can't both happen (one card: king AND queen), so P(E and F) = 0. **Independent** = E happening doesn't change P(F) (two coin flips). **Disjoint is NOT independent:** if they are disjoint, knowing E happened means F didn't.

The rules — spot the keyword

NOT: $P(E^c) = 1 - P(E)$ · **AT LEAST ONE:** $P(\text{at least 1}) = 1 - P(\text{none})$
OR: $P(E \text{ or } F) = P(E) + P(F) - P(E \text{ and } F)$ **(disjoint → the overlap is 0, just add)**
AND, independent: $P(E \text{ and } F) = P(E) \cdot P(F)$ · **AND, always:** $P(E \text{ and } F) = P(E) \cdot P(F \mid E)$
GIVEN: $P(F \mid E) = \frac{P(E \text{ and } F)}{P(E)} = \frac{N(E \text{ and } F)}{N(E)}$ · **Independent? check $P(F \mid E) = P(F)$: equal → independent, different → dependent**

"At least one": 1 − P(none). P(survive) = 0.9986: P(both of 2 survive) = 0.9986^2 = 0.9972; P(at least one of 1000 dies) = $1 - 0.9986^{1000}$ = 0.7536.

Counting — "how many ways"

- A **sequence of choices**: multiply the options at each step. 3-letter codes: 26·26·26 = 17,576; chair, vice-chair, secretary from 14: 14·13·12 = 2,184.
- **Permutation** = pick r of n, ORDER MATTERS (1st/2nd/3rd). ${}_nP_r = \frac{n!}{(n-r)!}$ ${}_7P_5$ = 7·6·5·4·3 = 2,520 (r numbers counting down); ${}_{10}P_3$ = 720; 7 in order: 7! = 5,040; note 0! = 1: ${}_5P_5$ = 5! = 120; ${}_nC_0 = {}_nC_n = 1$.
- **Combination** = pick r of n, order DOESN'T matter (committee, poker hand, sample). ${}_nC_r = \frac{n!}{r!\,(n-r)!}$ ${}_4C_2$ = 4!/(2!·2!) = 24/(2·2) = 6; ${}_{20}C_4$ = 4,845. Probability by counting: P(4 cards all the same rank) = 13 ranks ÷ ${}_{52}C_4$ = 13/270,725 ≈ 0.00005.

CH 5 — DISCRETE RANDOM VARIABLES ★Q2 ★Q3

- **Random variable X** = the number that comes out of a chance experiment. **Discrete** = you count it; its possible values are a list (x = 0, 1, 2, …). **Continuous** = you measure it; its possible values are an interval (time t ≥ 0, mass 0 < m ≤ 16 oz). **Probability distribution** = every value X can take with its probability.

VALID only if (1) every $0 \le P(x) \le 1$ AND (2) $\sum P(x) = 1$
$\mu_X = E(X) = \sum [x \cdot P(x)]$
$\sigma_X = \sqrt{\sum [(x-\mu)^2 \cdot P(x)]}$

Mean of a distribution = weighted average (each value counts as much as its probability) = the long-run average per trial. **SD** = typical distance from the mean, each squared gap weighted by its probability.

Q: "Insurance: premium $100, pays $10,000 if the client dies, P(dies) = 0.0014. Expected profit?" **A:** lives → +$100 with P = 0.9986; dies → 100 − 10,000 = −$9,900 with P = 0.0014. E(X) = 100(0.9986) − 9900(0.0014) = 99.86 − 13.86 = **$86** per policy → keep selling.

CH 5 (cont.) — BINOMIAL ★Q4

Binomial experiment = the same yes/no trial repeated: (1) FIXED number of trials n, (2) independent, (3) two outcomes (success / failure), (4) SAME p each time. X = number of successes. Not binomial if n isn't fixed ("until…") or picks without replacement are more than 5% of the group.

$P(x) = {}_nC_x\, p^x (1-p)^{n-x}$ ${}_nC_x = \frac{n!}{x!\,(n-x)!}$
$\mu = np$ $\sigma = \sqrt{np(1-p)}$
Unusual count (needs np(1 − p) ≥ 10): below μ − 2σ or above μ + 2σ · Excel: BINOM.DIST(x, n, p, FALSE) = P(x); TRUE = P(X ≤ x)

BINOMIAL — EVERY WAY THEY CAN ASK IT — worked (n = 10, p = 0.60, the ★Q4 numbers: P(0) = 0.0001, P(1) = 0.0016, P(2) = 0.0106, P(3) = 0.0425)
at most / at least / "or fewer" / "or more" keep the number IN. fewer than / more than / under / over leave it OUT. Short list → add the P(x)'s. Long list → 1 − (the short list on the other side).

exactly 3	$P(3) = {}_{10}C_3(0.6)^3(0.4)^7 = 120(0.216)(0.0016384)$ = **0.0425**
fewer than 3 = 2 or fewer (3 is OUT)	x = 0, 1, 2: $0.0001 + 0.0016 + 0.0106$ = **0.0123**
at most 3 = 3 or fewer (3 is IN)	x = 0, 1, 2, 3: $0.0001 + 0.0016 + 0.0106 + 0.0425$ = **0.0548**
more than 3 = 4 or more (3 is OUT)	x = 4 … 10 is long → $1 - P(0,1,2,3) = 1 - 0.0548$ = **0.9452**
at least 3 = 3 or more (3 is IN)	x = 3 … 10 is long → $1 - P(0,1,2) = 1 - 0.0123$ = **0.9877**
between 1 and 3 inclusive	x = 1, 2, 3: $0.0016 + 0.0106 + 0.0425$ = **0.0547**
between 1 and 3 NOT inclusive	only x = 2: P(2) = **0.0106**
at least 1	$1 - P(0) = 1 - 0.0001$ = **0.9999**
none	$P(0) = (0.4)^{10}$ = **0.0001**
all 10	$P(10) = (0.6)^{10}$ = **0.0060**
mean and SD for n = 200	$\mu = 200(0.6) = 120$; $\sigma = \sqrt{200(0.6)(0.4)} = \sqrt{48}$ = **6.93**
is 130 of 200 unusual?	usual range $\mu \pm 2\sigma = 120 \pm 13.9$ = 106.1 to 133.9 → 130 is inside → **not unusual**

Q: "Binomial or not? (1) Roll a pair of dice 10 times, count sums of 7. (2) Select flights until 10 late ones. (3) Pick 4 of 30 students, count females." **A:** (1) YES: n = 10, p = 6/36 (6 of the 36 sums are 7), x = 0, 1, …, 10 (x always runs 0 … n). (2) NO: n not fixed. (3) NO: no replacement, and 4 of 30 = 13% > 5% of the group → p changes.

CH 6 — CONTINUOUS & NORMAL

- A continuous variable has infinitely many values, so probability = **AREA under the curve** (pdf = the probability curve: never below 0, total area 1). **P(X = one exact value) = 0** so P(a < X < b) = P(a ≤ X ≤ b).

Uniform ★Q5

Uniform from a to b = a flat rectangle: every stretch of the same length is equally likely. Probability of an interval = its share of the total width.

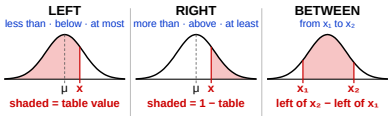
$\text{height} = \frac{1}{b-a}$ $P(c \le X \le d) = \frac{d-c}{b-a}$
value with share P below it: $x = a + P(b-a)$

Normal ★Q6 ★Q7

Normal = the bell curve: symmetric around μ, most values near the middle. μ slides it left/right, σ makes it wider or narrower.
Properties: symmetric about μ; mean = median = mode = μ (the peak); bends (inflection points) at μ ± σ; total area 1; tails never touch the axis; Empirical Rule: 68% of values within 1σ of μ, 95% within 2σ, 99.7% within 3σ (pieces 34 | 13.5 | 2.35 | 0.15 per side); half the area on each side of μ.

Standard normal Z: μ = 0, σ = 1. **The trick:** convert any value to a z-score and use ONE table (or the exam's hint): it gives the area to the LEFT of z.

$z = \frac{x-\mu}{\sigma}$ (round to 2 decimals) $x = \mu + z\sigma$
LEFT ("less than / below"): table value · RIGHT ("more than / above"): 1 − table value
BETWEEN: (left of the bigger z) − (left of the smaller z) · no −z row in the table? left of −z = 1 − (left of +z)



Steps: draw the bell, mark μ and x, shade → z = (x − μ)/σ to 2 decimals → hint or table (z = 1.67: row 1.6, column .07) = area to the LEFT → left: done · right: 1 − it · between: big left − small left.

z — EVERY WAY THEY CAN ASK IT — worked (the ★Q6 hints: left of −1.10 = 0.1357, of −0.60 = 0.2743, of 0.60 = 0.7257, of 1.45 = 0.9265)

left of −1.10	the hint is already the left area → **0.1357**
right of −1.10	$1 - 0.1357$ = **0.8643**
left of 1.45	the hint → **0.9265**
right of 1.45	$1 - 0.9265$ = **0.0735**
between −0.60 and 1.45	$0.9265 - 0.2743$ = **0.6522**
between −1.10 and −0.60 (both negative)	the bigger z is −0.60: $0.2743 - 0.1357$ = **0.1386**
between 0.60 and 1.45 (both positive)	$0.9265 - 0.7257$ = **0.2008**
outside −1.10 and 1.45 (two tails)	$0.1357 + (1 - 0.9265) = 0.1357 + 0.0735$ = **0.2092**
z with area 0.10 to its LEFT	find the hint closest to 0.10 → z = **−1.28**
z with area 0.025 to its RIGHT	left area = $1 - 0.025 = 0.975$ → z = **1.96**
two z's cutting the middle 90%	0.05 in each tail → left areas 0.05 and 0.95 (0.95 is halfway between 1.64 and 1.65) → z = **±1.645**
N(80, 10): P(70 < X < 95)	$z = \frac{70-80}{10} = -1.00 \to 0.1587$; $z = \frac{95-80}{10} = 1.50 \to 0.9332$ → $0.9332 - 0.1587$ = **0.7745**
N(80, 10): P(X > 95)	z = 1.50 → 0.9332 → $1 - 0.9332$ = **0.0668**
N(80, 10): P(X ≤ 70)	z = −1.00 → left area = **0.1587**

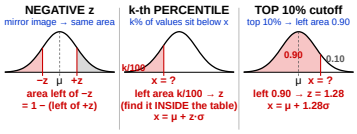
USING THE z HINTS ON THE EXAM (no table needed)

Every z you need is printed as a hint; a hint is a table value. Still compute your own z to 2 decimals and match it to a hint — no match means your arithmetic is wrong.

"P(Z < 1.67) = 0.9525" / "area to the left of 1.67 is 0.9525" → the LEFT area: below → write it · above → 1 − it · between → bigger − smaller
"P(Z > 1.67) = 0.0475" (a right tail) → that IS the "above" answer; "below" = 1 − 0.0475
"z = 2.05 for the 98th percentile" / "z = 1.28 for the top 10%" → x = μ + z·σ; bottom 10% → negative z (z = −1.28 → x = μ − 1.28σ)
"P(Z < −1.43) = 0.0764" is already a LEFT area — write it as is · "is it unusual?" has no hint: |z| > 2 (x beyond μ ± 2σ) → unusual

Backwards — find x from a percent / percentile ★Q7(d)

The area is GIVEN: words → area to the LEFT ("top 10%" → 0.90, "bottom 5%" → 0.05, "k-th percentile" → k/100, "middle 80%" → 0.10 and 0.90) → the hint's z (or the closest table entry) → x = μ + zσ.



NORMAL — EVERY WAY THEY CAN ASK IT — worked (IQ: μ = 100, σ = 15, the ★Q7 numbers)

above 125 (same as at least 125)	$z = \frac{125-100}{15} = 1.67$ → hint 0.9525 → $1 - 0.9525$ = **0.0475**
below 90 (same as at most 90)	$z = \frac{90-100}{15} = -0.67$ → hint 0.2514 → already the left area → **0.2514**
between 110 and 140	$z = \frac{110-100}{15} = 0.67 \to 0.7486$; $z = \frac{140-100}{15} = 2.67 \to 0.9962$ → $0.9962 - 0.7486$ = **0.2476**
outside 70 to 130 (below 70 OR above 130)	$z = -2.00 \to 0.0228$; $z = 2.00 \to 0.9772$ → $0.0228 + (1 - 0.9772)$ = **0.0456**
exactly 100	one value has no area → **0.0000**
98th percentile (98% below)	left area 0.98 → z = 2.05 → $x = 100 + 2.05(15) = 100 + 30.75$ = **130.75**
top 5% cutoff	95% below it → left area 0.95 → z = 1.645 → $x = 100 + 1.645(15) = 124.675 \approx$ **124.7**
bottom 10% cutoff	left area 0.10 → z = −1.28 → $x = 100 - 1.28(15)$ = **80.8**
middle 95% (two cutoffs)	2.5% in each tail → left areas 0.025 and 0.975 → z = ±1.96 → $100 \pm 1.96(15) = 100 \pm 29.4$ → **70.6 to 129.4**
the value with 30% above it	70% below it → left area 0.70 → z = 0.52 → $x = 100 + 0.52(15)$ = **107.8**
is a score of 65 unusual?	$z = \frac{65-100}{15} = -2.33$ → tail area 0.0099 < 0.05 (and |z| > 2) → **yes, unusual**
interpret "area above 125 = 0.0475"	(1) 4.75% of children score above 125; (2) one random child has a 0.0475 chance of scoring above 125

Percentile → z: 1st −2.33 · 5th −1.645 · 10th −1.28 · 20th −0.84 · 25th −0.67 · 75th 0.67 · 80th 0.84 · 90th 1.28 · 95th 1.645 · 97.5th 1.96 · 98th 2.05 · 99th 2.33 · middle 80 / 90 / 95 / 99% ↔ ±1.28 / ±1.645 / ±1.96 / ±2.575

- **Is it normal?** histogram bell-shaped, one peak, symmetric, few outliers → yes; a **normal probability plot** (observed values vs the z-scores expected if normal) close to a straight line → yes; a long tail or a bent plot → not normal.

WRITE THIS SENTENCE — answer templates

- **Interpret the mean / expected value:** "Over many [series], the average [games per series] is about [5.48]." Insurance: "$86 profit per policy on average → positive → keep selling."
- **Valid distribution? (Q2):** "Not valid: the probabilities add to 0.73, not 1" / "not valid: P(2) = −0.1 is negative" / "each P(x) is 0–1 and they add to 1 → valid."
- **Unusual?** "P = 0.025 < 0.05 → unusual" / "count 130 is inside μ ± 2σ = 106.1 to 133.9 → not unusual." **Independent?** "P(A | B) = 0.2381 ≠ P(A) = 0.2051 → dependent." **Sketch:** uniform = "flat line at height 1/30 from 70 to 100, 0 elsewhere"; normal = bell at μ, ticks μ ± σ, 2σ, 3σ, shade, write the area.
- **Don't lose points:** "or" → subtract the overlap once · "and" → the one cell · "given" → divide by the given group · binomial σ needs the √ · uniform never uses z · normal: z to 2 decimals, hint = area to the LEFT, "above" → 1 − left · 4 decimals and the formula every time.

Calculator: 36 ÷ 78 = 0.4615 · 0.4 ^ 7 = 0.0016384 · 10 [nCr] 3 = 120 (TI-84: MATH → PRB; TI-89: 2nd MATH → Probability → nCr(10,3); no key: 10! ÷ (3!·7!)) · √48 = 6.93 · keep decimals (round only z, to 2).
Order (minutes): Q1 6 · Q2 2 · Q5 5 · Q6 6 · Q7 9 · Q3 10 · Q4 10. Formula first, box every answer.